PART 4

Maintenance History File

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All entries below are synthetic. Asset: PLC-FIL-02, CPU 1214C DC/DC/DC, firmware V3.0 baseline.

1. Asset register extract

Tag	Description	Installed	Notes
PLC-FIL-02	CPU 1214C DC/DC/DC	14 Mar 2024	Firmware V3.0; access protection enabled
SM-FIL-02-1	SM 1223 DI/DQ 16x24VDC	14 Mar 2024	Expansion module slot 2
SM-FIL-02-2	SM 1231 AI 4 x 16 bit	14 Mar 2024	Fill level and pressure sensors
BB-FIL-02	BB 1297 battery board	05 Jun 2025	Added for RTC backup
MC-FIL-02	SIMATIC memory card	14 Mar 2024	Program card, spare on shelf

2. Work history

Date	WO #	Type	Description	Performed by	Outcome
14 Mar 2024	WO-24-0112	Install	Commissioning of CPU, SM 1223 and SM 1231; project downloaded; baseline archive made	A. Rao	Closed
22 Sep 2024	WO-24-0987	Inspection	Level C annual inspection; project archive and CPU upload taken	R. Khan, A. Rao	No findings
09 Jan 2025	WO-25-0034	Corrective	Intermittent HMI comms loss traced to damaged Ethernet patch cable; cable replaced, link tested	R. Khan	Closed (S3)
05 Jun 2025	WO-25-0521	Modification	BB 1297 battery board fitted into signal board slot to extend RTC backup	A. Rao	Closed
19 Sep 2025	WO-25-1180	Inspection	Level C annual inspection; diagnostic buffer reviewed, 3 historical analog wire-break events explained	R. Khan, A. Rao	Minor finding
03 Dec 2025	WO-25-1402	Corrective	Analog input channel 2 on SM 1231 drifting; loose terminal re-torqued, scaling re-verified	R. Khan	Closed (S3)
17 Mar 2026	WO-26-0310	Inspection	Level B quarterly check; cabinet filters cleaned; cycle time verified	R. Khan	OK
28 Jun 2026	WO-26-0742	Incident	S2: CPU ERROR LED flashing after memory card removed by mistake during cabinet work; card reseated, CPU restarted, diagnostic buffer exported	R. Khan, A. Rao	Closed; training added
25 Sep 2026	WO-26-1066	Inspection	Level C annual inspection; full archive taken; compare online/offline identical; spares reviewed	R. Khan, A. Rao	No findings
29 Oct 2026	WO-26-1190	Preventive	Planned PSU and fan check in control cabinet; all within limits	R. Khan	OK

3. Spare parts status

Part	Stock	Min	Status
CPU 1214C DC/DC/DC	1	1	OK
SM 1223 DI/DQ 16x24VDC	1	1	OK
SM 1231 AI 4 x 16 bit	0	1	**Reorder**

Part	Stock	Min	Status
SIMATIC memory card	2	1	OK
24 V DC power supply	1	1	OK

4. Open items

- Order one SM 1231 AI 4 x 16 bit spare (due 15 Nov 2026).
- Next Level B quarterly check due 17 Dec 2026.
- Review password vault entry at next Level C.